Activity No. 13	
Working with Files and Directories in Linux	
Course Code: CPE 201A	**Program:** BSCPE
Course Title: COMPUTER SYSTEM ADMINISTRATION AND TROUBLESHOOTING	**Date Performed:** 10/20/2026
Section: CPE11S2	**Date Submitted:** 10/20/2026
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1. Objective/s:

This activity aims to navigate and manage files and directories using command line interface.

2. Intended Learning Outcome/s:

The students should be able to:

2.1 Demonstrate how to navigate home and system directories
2.2 Demonstrate how to how to list files and directories.
2.3 Demonstrate how to use globbing to manipulate files and directories.
2.4 Demonstrate how to create, move and delete files and directories.

3. Discussion:

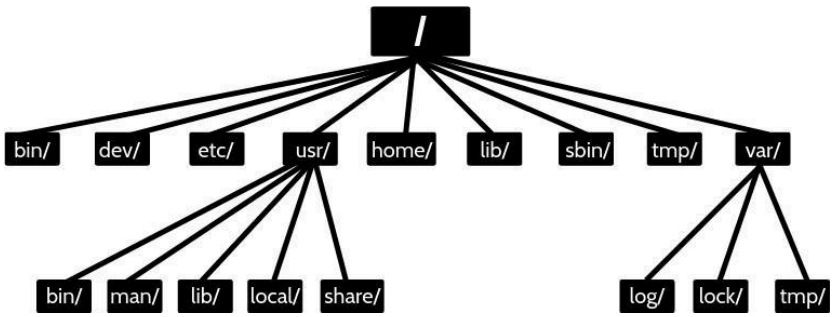
Files and Directories

The Linux filesystem is similar to other operating system's filesystems in that it contains *files* and *directories*. Files contain data such as human-readable text, executable programs, or binary data that is used by the computer.

Directories are used to create organization within the filesystem. Directories can contain files and other directories.

Directory Structure

On a Windows system, the *top level* of the directory structure is called ***My Computer*.** The Linux directory structure, called a ***filesystem***, also has a top level called the ***root directory*** (symbolized by the **slash /character**).



Home Directory

On most Linux distributions there is a directory called **home** under the root /directory. Under this **/home** directory

there is a directory for each user on the system. When a user opens a shell, they should automatically be placed in their home directory. The user has the full control to create and delete additional files and directories in their home directory. Most other directories in a Linux filesystem are protected with *file permissions*. The home directory has a special symbol used to represent it, the **tilde ~ character**.

The directory name is the same as the name of the user. So, a user named sysadmin would have a *home directory* called /home/sysadmin:

File and Directory Names

File and directory names in Linux can contain lower case and upper case letters, numbers, spaces and special characters. However, since many special characters have a special meaning in the Linux shell, it is good practice to not use spaces or special characters when naming files or directories. Spaces, for example, need the *escape character* \ to be entered correctly

Paths

A path is a list of directories separated by the / character. There are two types of paths: *absolute* and *relative*. For example, /home/sysadmin is a path to the home directory

A. Absolute Paths

Absolute paths allow the user to specify the exact location of a directory. Absolute paths always starts at the root directory, and therefore it always begins with the / character.

B. Relative Paths

A relative path gives directions to a file relative to the current location in the filesystem. The user must currently be in a directory that contains objects in the path. Relative paths start with the name of a directory.

Special Relative Paths

The single period . character always represents the current directory.
Two period .. characters always represents one directory higher relative to the current directory, sometimes referred to as the parent directory.

4. Resources:

Personal Computer with installed Virtual Box
Ubuntu Server or Desktop virtual machine

5. Procedure:

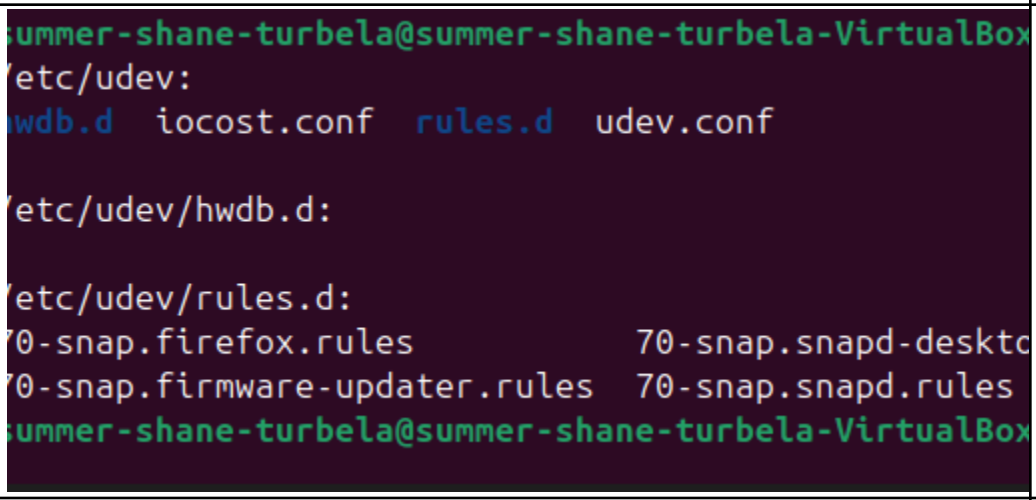
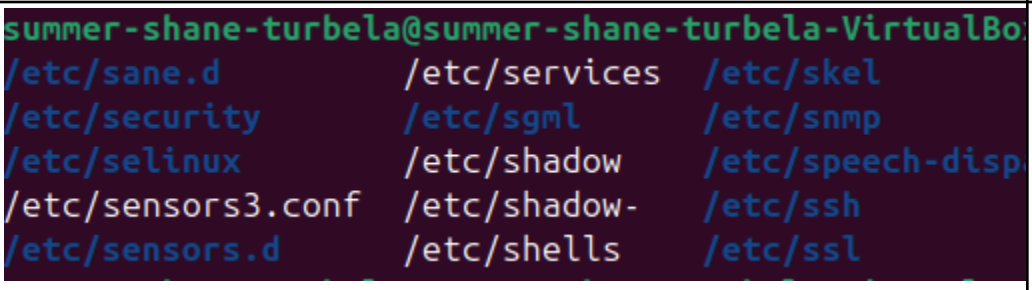
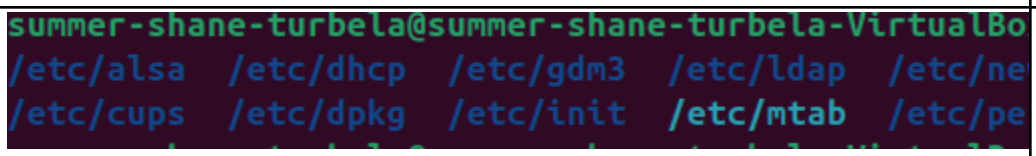
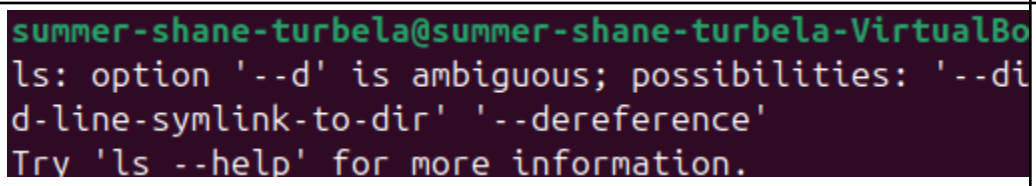
1. Login using your username and password.
2. Use terminal emulator application (if you are using desktop version)
3. Execute the following commands to navigate files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

Command	Screenshot	Explanation
1. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ pwd /home/summer-shane-turbela	current status of

		your terminal cmds
2. cd /	summer-shane-turbela@summer-shane-turbela-VirtualBox:~$	changing to root directory
3. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo	now there's a "/" in every line
4. cd	summer-shane-turbela@summer-shane-turbela-VirtualBo	goes back to the former directory earlier
5. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /home/summer-shane-turbela	checks if it's currently in the home directory
6. cd /home	summer-shane-turbela@summer-shane-turbela-VirtualBo	moves you to the /home directory
7. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /home	checks if the current line does show /home directory
8. cd ~	summer-shane-turbela@summer-shane-turbela-VirtualBo	a shortcut command that takes you to your home directory
9. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /home/summer-shane-turbela	verifies if current line shows home/sysadmin
10. echo ~ ~sysadmin ~root ~mail ~nob	summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ echo ~ /home/summer-shane-turbela summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ ~summer-shane-turbela ~root ~mail ~nobody bash: /home/summer-shane-turbela: Is a directory	echo ~ replaces sysadmin as ur tilde with home personal folder(/home/sysadmin), root is the

ody where sysadmin is your username		superuser, ~mail is a background system acc, and ~nobody is not a true directory
11. cd ~root	summer-shane-turbela@summer-shane-turbela-VirtualBo: bash: cd: /root: Permission denied	cmd that's only for administrators, users cannot simply access this
12. cd /usr/bin	summer-shane-turbela@summer-shane-turbela-VirtualBo:	cmd that moves into the Linux filesystem
13. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /usr/bin	verifies if we're inside the /usr/bin
14. cd /usr	summer-shane-turbela@summer-shane-turbela-VirtualBo:	moves into the user resource directory
15. type ls	summer-shane-turbela@summer-shane-turbela-VirtualBo ls is aliased to `ls --color=auto'	outputs what the commands is
16. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /usr	outputs that we're in the /usr directory
17. cd /usr/share/doc	summer-shane-turbela@summer-shane-turbela-VirtualBo	cmd where you move to the system's documentation directory
18. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /usr/share/doc	outputs what current directory are we in
19. cd bash	summer-shane-turbela@summer-shane-turbela-VirtualBo	a cmd where you move into

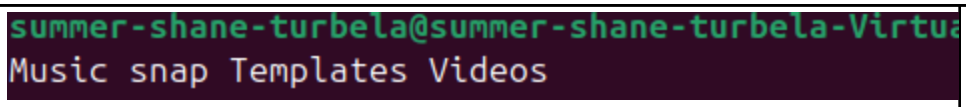
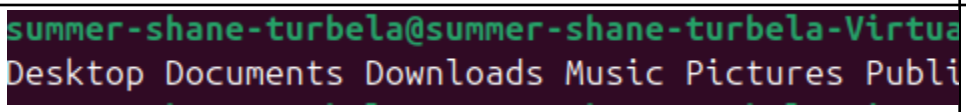
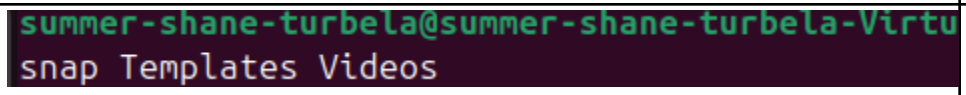
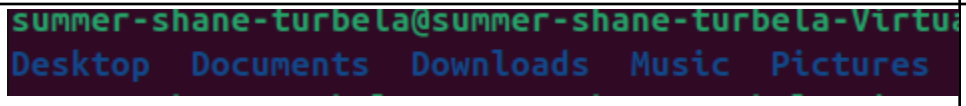
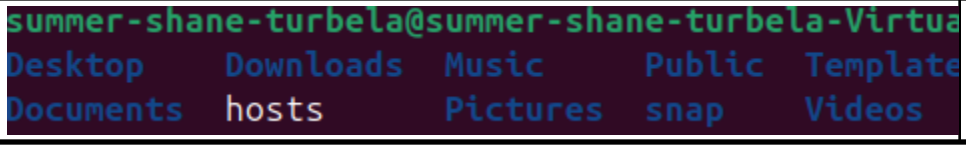
		the bash shell
20. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /usr/share/doc/bash	now it shows we're currently in the bash shell
21. cd ..	summer-shane-turbela@summer-shane-turbela-VirtualBox	a cmd that works as a backspace
22. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /usr/share/doc	now you're back to the directory before the bash cmd
23. cd ../dict	summer-shane-turbela@summer-shane-turbela-VirtualBo	moves you to the directory named dict
24. pwd	summer-shane-turbela@summer-shane-turbela-VirtualBo /usr/share/dict	now verifies that we're inside the directory dict
25. cd	summer-shane-turbela@summer-shane-turbela-VirtualBo	returns to home directory
26. ls	summer-shane-turbela@summer-shane-turbela-VirtualBo Desktop Documents Downloads Music Pictures Pub	ls is the list of the directories at home
27. ls -a	summer-shane-turbela@summer-shane-turbela-VirtualBo . .bashrc Desktop .gnupg Pictures .. .cache Documents .local .profile .bash_logout .config Downloads Music Public	reveals hidden stuff in the home directory
28. ls -l /etc/hosts	summer-shane-turbela@summer-shane-turbela-VirtualBox -rw-r--r-- 1 root root 246 Oct 13 11:43 /etc/hosts	cmd that displays the path, permissions, roots, the no. of hard links to the file, the size of bytes, and when

		was it last modified
29. ls -R /etc/udev		shows all the files inside the udev and the subcategories in it
30. ls -d /etc/s*		searches the /etc directory for any file folder (only for the ones that starts with the letter “s”)
31. ls -d /etc/????		commands the system to list directories/files inside /etc that’s four characters long
32. ls –d /etc/[abcd]*		looks inside the /etc and if there’s any matching file that starts with a,b,c, or d

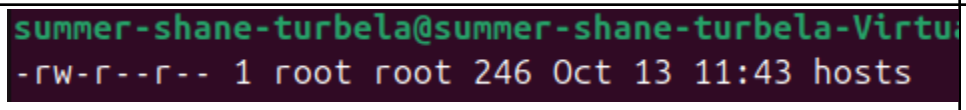
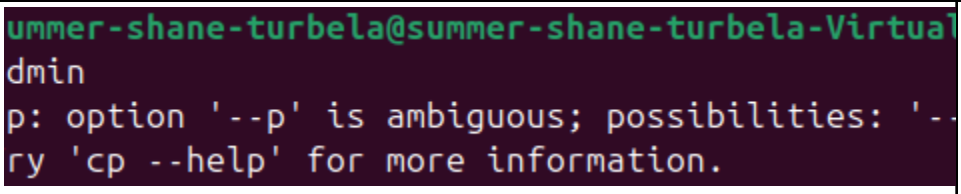
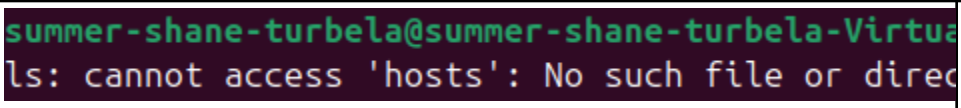
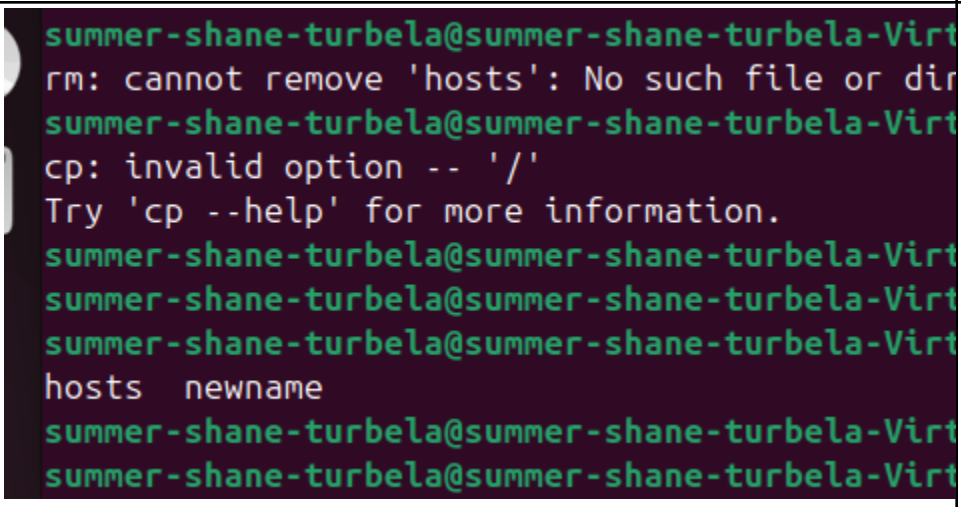
4. Execute the following commands to manage files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

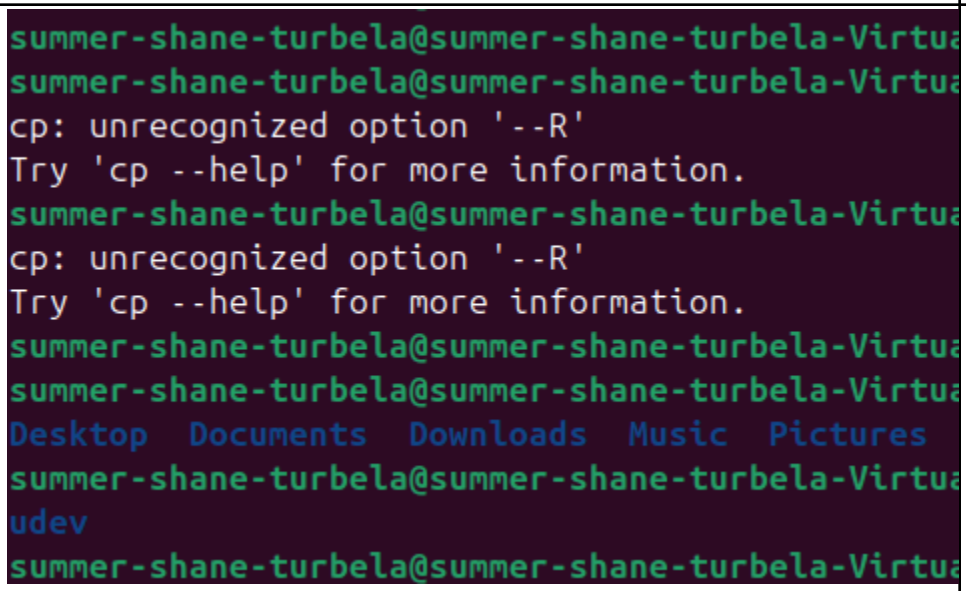
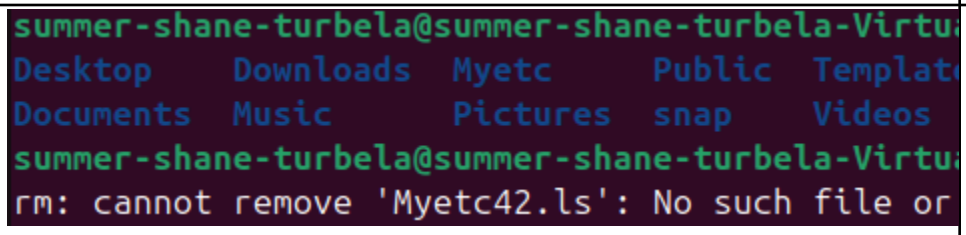
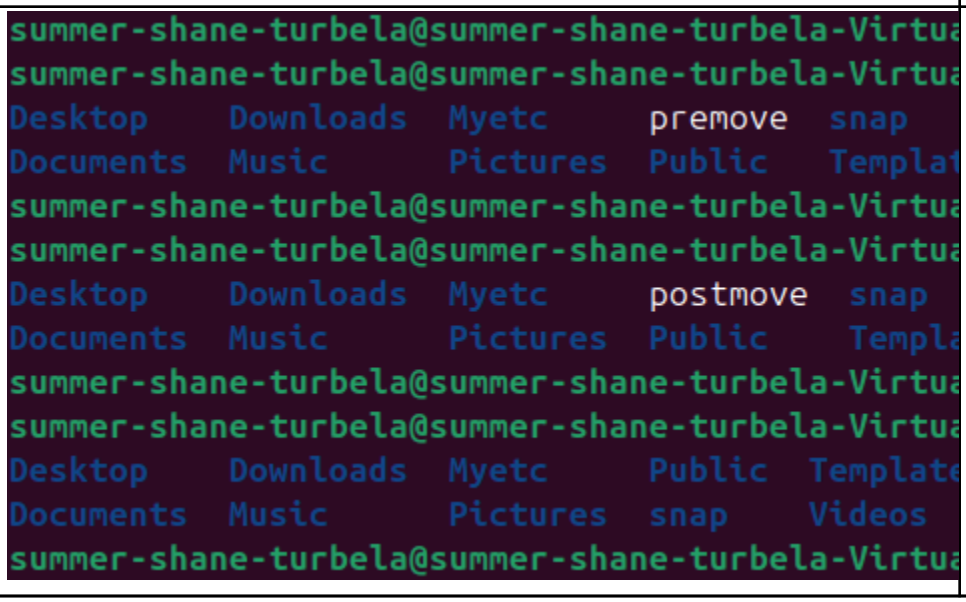
Command	Screenshot	Explanation
1. echo *	summer-shane-turbela@summer-shane-turbela-Virtua Desktop Documents Downloads Music Pictures Publi	list all the files and directory inside the working directory
2. echo D*	summer-shane-turbela@summer-shane-turbela-Virtua Desktop Documents Downloads	displays files and directory inside the working directory whose folder name starts with the letter “D”
3. echo P*	summer-shane-turbela@summer-shane-turbela-Virtua Pictures Public	displays files and directory inside the working directory whose folder name starts with the letter “P”
4. echo *s	summer-shane-turbela@summer-shane-turbela-Virtua Documents Downloads Pictures Templates Videos	displays files and directory inside the working directory whose folder name ends with the lowercase letter “s”
5. echo D*n*s	summer-shane-turbela@summer-shane-turbela-Virtua Documents Downloads	displays files and directory inside the working directory whose folder name starts with the letter “D”, contains the lowercase

		letter “n”, and ends with the lowercase “s”
6. echo ??????	summer-shane-turbela@summer-shane-turbela-Virtua Public Videos	displays files and directory inside the working directory whose folder name are 6 characters long
7. echo D???????	ummer-shane-turbela@summer-shane-turbela-Virtual ocuments Downloads	displays files and directory inside the working directory whose folder name starts with the letter “D” and is 9 characters long
8. echo ?????*s	summer-shane-turbela@summer-shane-turbela-Virtua Documents Downloads Pictures Templates Videos	displays files and directory inside the working directory whose folder name ends with the lowercase letter “s” and is 6 characters long
9. echo [DP]*	summer-shane-turbela@summer-shane-turbela-Virtua Desktop Documents Downloads Pictures Public	displays files and directory inside the working directory whose folder name that’s either D or or P

10. echo [!DP]*		displays files and directory inside the working directory whose folder name that doesn't start with either D or or P
11. echo [D-P]*		displays files and directory inside the working directory whose folder name that start with any capital letters from D through P
12. echo [!D-P]*		displays files and directory inside the working directory whose folder name that doesn't start with any capital letters from D through P
13. ls		ls is the list of the directories at home
14. cp /etc/hosts hosts		Copies /etc/hosts file and names the new one hosts
15. ls		Verifies if the command worked and it

		successfully created
16. rm hosts	summer-shane-turbela@summer-shane-turbela-Virtua	Deletes the newly made copy (hosts)
17. ls	summer-shane-turbela@summer-shane-turbela-Virtua Desktop Documents Downloads Music Pictures	Verifies if the copy is gone
18. cp –v /etc/hosts hosts	summer-shane-turbela@summer-shane-turbela-Virtua '/etc/hosts' -> 'hosts'	copies and turns into verbose mode (displays confirmation message of what action was taken)
19. ls	summer-shane-turbela@summer-shane-turbela-Virtua Desktop Downloads Music Public Template Documents hosts Pictures snap Videos	verifies if the copy is there
20. rm hosts	summer-shane-turbela@summer-shane-turbela-Virtu	Deletes the newly made copy
21. ls	summer-shane-turbela@summer-shane-turbela-V Desktop Documents Downloads Music Pictu	verifies if the copy has been removed
22. cp –v /etc/hosts .	summer-shane-turbela@summer-shane-turbela-Virtua '/etc/hosts' -> './hosts'	Copies both /etc/hosts and the hosts while keeping the same name from a directory to your current directory
23. ls	summer-shane-turbela@summer-shane-turbela-Virtua Desktop Downloads Music Public Templat Documents hosts Pictures snap Videos	Verifies again if it's there
24. rm hosts	summer-shane-turbela@summer-shane-turbela-Virtua	Deletes the copy
25. ls	summer-shane-turbela@summer-shane-turbela-Virtua Desktop Documents Downloads Music Pictures	Verifies if the copy is gone

26. cd /etc		Moves into /etc folder
27. ls -l hosts		displays detailed information about the host file
28. cp –p hosts/home/sysadmin		Errors because it interprets of copying the files inside the hosts that has home inside and sysadmin inside home
29. cd		errors because it's a file not a directory
30. ls –l hosts		Displays the detailed information of the file hosts
31. rm hosts 32. cp -p /etc/hosts ~ 33. cp hosts newname 34. ls –l hosts newname 35. rm hosts newname		31. Deletes the file 32. Restores the file and copies original to the current folder 33. Duplicates hosts into a new file named newname 34. Shows the detailed information of both files 35. Deletes the files named hosts

		and newname
36. mkdir Myetc 37. cp –R /etc/udev Myetc 38. ls –l Myetc 39. ls –lR Myetc		36. makes a directory named Myetc 37.recursivey copies the contents of the udev folder to Myetc 38. Shows what's inside Myetc 39. Recursively list what's inside Myetc
40. ls 41. rm -r Myetc 42. Ls		40 verifies the contents of the directory 41. Deletes the directory Myetc including its contents 42. confirms that the folder is gone
43. touch premove 44. ls 45. mv premove postmove 46. ls 47. rm postmove		43. Command that creates a file named premoove 44. Verifies if the file exists 45. renames and moves the file 46. confirm if the new file name exists 47. Deletes the renamed file

6. Supplementary Activity:

Copy screen shot(s) of the following tasks:

1. Create a directory CPE201 in the Documents directory.

```
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ mkdir Documents/CPE201
```

2. Create a directory <lastname> and <firstname> under CPE201 directory.

```
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ mkdir Documents/CPE201/T
urbela Documents/CPE201/Summer-Shane
```

3. Create two text files: (1) A.txt and (2) B.txt in the <lastname> directory.

```
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ touch Documents/CPE201/
urbela/A.txt Documents/CPE201/Turbela/B.txt
```

4. Copy the content of <lastname> directory to <firstname> directory.

```
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ cp -r Documents/CPE201/T
urbela/. Documents/CPE201/Summer-Shane/
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ ls Documents/CPE201/Summ
er-Shane*
A.txt  B.txt
```

5. Delete the content of <lastname> directory.

```
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ rm -rf Documents/CPE201/
Turbela/*
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ ls Documents/CPE201/Turb
ela
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$
```

6. Create a directory CPE201_backup in the Documents directory.

```
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ mkdir Documents/CPE201_backup
```

7. Copy the content of CPE201 to CPE201_backup directory.

```
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ cp -r Documents/CPE201/. Documents/CPE201_backup
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ ls Documents/CPE201
Summer-Shane  Turbela
```

8. Remove the CPE231 directory.

```
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ rm -r Documents/CPE201
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$ ls Documents
CPE201_backup
summer-shane-turbela@summer-shane-turbela-VirtualBox:~$
```

7. Conclusion:

Easy but confusing, I have to follow through every step carefully. I made some small mistakes that doesn't affect the needed output.

8. Assessment (Rubric for Laboratory Performance):

TIP Rubric E (1) (1) (1)
Criteria
Performance Indicators 1. Apply appropriate techniques, skills, and modern tools to perform a c engineering task.
Performance Indicators 2. Demonstrate skills in applying different techniques and modern tools engineering problems.1. Apply appropriate techniques, skills, and modern a discipline-specific engineering task.
Performance Indicators 3. Recognize the benefits and constraints of modern engineering tools.De in applying different techniques and modern tools to solve engineering p appropriate techniques, skills, and modern tools to perform a discipline- task.